

Q.No:6

Question description

There are n boys and m girls in a school. Next week a school dance will be organized. A dancing pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Constraints

- $1 \leq n, m \leq 500$
- $1 \leq k \leq 1000$
- $1 \leq a \leq n$
- $1 \leq b \leq m$

Input

The first input line has three integers n , m and k : the number of boys, girls, and potential pairs. The boys are numbered $1, 2, \dots, n$, and the girls are numbered $1, 2, \dots, m$.

After this, there are k lines describing the potential pairs. Each line has two integers a and b : boy a and girl b are willing to dance together.

Output

First print one integer rr : the maximum number of dance pairs. After this, print lines describing the pairs. You can print any valid solution.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 1
3 1
```

EXPECTED OUTPUT

```
2
1 2
2 1
```

Test Case 2

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 3
3 2
```

EXPECTED OUTPUT

```
3
1 1
2 3
3 2
```

Mandatory Test Cases

Code:

```
#include <stdio.h>
#include <string.h>

#define MAX 505
#define INF 1000000000

int boys, girls, k;
int adj[MAX][MAX];
int adjCount[MAX];
```

```
int pairBoy[MAX];
int pairGirl[MAX];
int dist[MAX];
void link(int u, int v)
{
adj[u][adjCount[u]++] = v;
}

int bfs(int n)
{
int queue[MAX];
int front = 0, rear = 0;
int i, u, v;
int found = 0;

for (i = 1; i <= n; i++)
{
if (pairBoy[i] == 0)
{
dist[i] = 0;
queue[rear++] = i;
}
else
{
dist[i] = INF;
}
}

while (front < rear)
{
u = queue[front++];

for (i = 0; i < adjCount[u]; i++)
{
v = adj[u][i];

if (pairGirl[v] == 0)
{
found = 1;
}
else if (dist[pairGirl[v]] == INF)
{
dist[pairGirl[v]] = dist[u] + 1;
queue[rear++] = pairGirl[v];
}
}
}

return found;
}

int dfs(int u)
{
int i, v;
```

```
for (i = 0; i < adjCount[u]; i++)
{
    v = adj[u][i];

    if (pairGirl[v] == 0 ||
        (dist[pairGirl[v]] == dist[u] + 1 &&
         dfs(pairGirl[v])))
    {
        pairBoy[u] = v;
        pairGirl[v] = u;

        return 1;
    }
}

dist[u] = INF;

return 0;
}

int main()
{
    int i;
    int u, v;
    int answer = 0;

    scanf("%d %d %d", &boys, &girls, &k);

    memset(adjCount, 0, sizeof(adjCount));
    memset(pairBoy, 0, sizeof(pairBoy));
    memset(pairGirl, 0, sizeof(pairGirl));

    for (i = 0; i < k; i++)
    {
        scanf("%d %d", &u, &v);
        link(u, v);
    }

    while (bfs(boys))
    {
        for (i = 1; i <= boys; i++)
        {
            if (pairBoy[i] == 0)
            {
                if (dfs(i))
                {
                    answer++;
                }
            }
        }
    }

    printf("Output:\n");
    printf("%d\n", answer);
}
```

```
for (i = 1; i <= boys; i++)
{
    if (pairBoy[i] != 0)
    {
        printf("%d %d\n", i, pairBoy[i]);
    }
}

return 0;
}
```

Output:

Test_case1:

```
● aadityajaiswal@Aadit
3 2 4
1 1
1 2
2 1
3 1
Output:
2
1 2
2 1
● aadityajaiswal@Aadit
```

Test_case2:

```
● aadityajaiswal@Aad
3 2 4
1 1
1 2
2 3
3 2
Output:
3
1 1
2 3
3 2
○ aadityajaiswal@Aad
```